

DOKUMENTACJA GAIA

1. Opis i Wymagania

Właściciele zwierząt domowych, ze szczególnym uwzględnieniem psów, na co dzień zmagają się z licznymi obowiązkami profilaktycznymi i opiekuńczymi, takimi jak szczepienia, odrobaczanie czy ochrona przeciwpasożytnicza. Wiele z tych zadań sprawia, że opiekunowie często gubią się w harmonogramach. Powąжным wyzwaniem jest również trudno w prawidłowym rozpoznawaniu sygnałów świadczących o pogorszeniu samopoczucia lub chorobie pupila. Brak stałego monitoringu stanu i zachowania zwierzęcia, zwłaszcza podczas nieobecności właściciela w domu, prowadzi do niepokoju i opóźnień w reakcji na potencjalne zagrożenia zdrowotne. Ponadto trudno w interpretacji objawów skutkują skrajnymi postawami: bagatelizowaniem problemów z powodu braku wiadomości lub nadmiernymi wizytami u weterynarza, co generuje nieuzasadnione koszty i niepotrzebnie obciąża kliniki weterynaryjne.

- Pomiar funkcji życiowych zwierzęcia
- Zapisywanie położenia zwierzęcia
- Kontrola zachowania zwierzęcia
- Monitorowanie stanu zdrowia
- Przypominanie o wizytach u weterynarza
- Umawianie wizyt u weterynarza

2. Model Ról

Role: PetOwner

Description	Entity responsible for the care and well-being of a pet, including scheduling veterinary appointments and monitoring health indicators.
Responsibilities	Liveness: (ReviewPhysiologicalData . ReviewLocationData . ReviewBehaviorData . ReceiveHealthAlert . ScheduleAppointment . ReceiveVetReminder)^ω Safety: UniqueID(Pet) != Null PhysiologicalData != Null LocationData != Null BehaviorData != Null HealthStatus != Null AlertData != Null AppointmentRequest != Null ReminderData != Null AppointmentConfirmation != Null
Permissions	READ: PhysiologicalData, LocationData, BehaviorData, HealthStatus, AlertData, AppointmentRequest, ReminderData WRITE: AppointmentConfirmation CONSUME: PhysiologicalData, LocationData, BehaviorData, HealthStatus, AlertData, AppointmentRequest, ReminderData PRODUCE: AppointmentConfirmation
Activities	ReviewPhysiologicalData, ReviewLocationData, ReviewBehaviorData, ReceiveHealthAlert, ScheduleAppointment, ReceiveVetReminder
Protocols	RecordPhysiologicalData, RecordLocation, AnalyzeBehavior, SendHealthAlert, ScheduleAppointment, SendVetReminder

Role: Pet

Description	The animal being monitored, whose health, behavior, and location data are tracked by the system.
Responsibilities	Liveness: [] Safety: UniqueID(Pet) != Null PhysiologicalData != Null LocationData != Null BehaviorData != Null HealthStatus != Null AlertData != Null AppointmentRequest != Null ReminderData != Null
Permissions	READ: PhysiologicalData, LocationData, BehaviorData, HealthStatus, AlertData, AppointmentRequest, ReminderData WRITE: PhysiologicalData, LocationData, BehaviorData, HealthStatus, AlertData, AppointmentRequest, ReminderData CONSUME: PhysiologicalData, LocationData, BehaviorData, HealthStatus, AlertData, AppointmentRequest, ReminderData PRODUCE: PhysiologicalData, LocationData, BehaviorData, HealthStatus, AlertData, AppointmentRequest, ReminderData
Activities	
Protocols	

Role: VeterinaryClinic

Description	Entity that provides veterinary services, including consultations, treatments, and health monitoring for pets.
Responsibilities	Liveness: (ConsultPetHealth . ProvideTreatment)^ω Safety: UniqueID(Pet) != Null HealthStatus != Null
Permissions	READ: HealthStatus WRITE: HealthStatus CONSUME: HealthStatus PRODUCE: HealthStatus
Activities	ConsultPetHealth, ProvideTreatment
Protocols	MonitorHealthStatus

Role: HealthMonitor	
Description	System component responsible for collecting and analyzing physiological data from the pet to detect health anomalies.
Responsibilities	Liveness: (CollectPhysiologicalData . AnalyzePhysiologicalData . DetectHealthAnomaly . RecordPhysiologicalData . MonitorHealthStatus . SendHealthAlert)^ω Safety: UniqueID(Pet) != Null PhysiologicalData != Null HealthStatus != Null AlertData != Null
Permissions	READ: PhysiologicalData WRITE: PhysiologicalData, HealthStatus, AlertData CONSUME: PhysiologicalData PRODUCE: PhysiologicalData, HealthStatus, AlertData
Activities	CollectPhysiologicalData, AnalyzePhysiologicalData, DetectHealthAnomaly
Protocols	RecordPhysiologicalData, MonitorHealthStatus, SendHealthAlert

Role: BehavioralAnalyzer	
Description	System component that evaluates pet behavior patterns to identify potential issues or changes in well-being.
Responsibilities	Liveness: (CollectBehaviorData . AnalyzeBehaviorPattern . AnalyzeBehavior)^ω Safety: UniqueID(Pet) != Null BehaviorData != Null
Permissions	READ: BehaviorData WRITE: BehaviorData CONSUME: BehaviorData PRODUCE: BehaviorData
Activities	CollectBehaviorData, AnalyzeBehaviorPattern
Protocols	AnalyzeBehavior

Role: LocationTracker	
Description	System component that records and reports the geographical position of the pet at regular intervals.
Responsibilities	Liveness: (CollectLocationData . RecordLocation)^ω Safety: UniqueID(Pet) != Null LocationData != Null
Permissions	READ: LocationData WRITE: LocationData CONSUME: LocationData PRODUCE: LocationData
Activities	CollectLocationData
Protocols	RecordLocation

Role: AppointmentScheduler	
Description	System component that manages scheduling and reminders for veterinary appointments based on pet's health records.
Responsibilities	Liveness: (ProcessAppointmentRequest . GenerateReminder . ScheduleAppointment . SendVetReminder)^ω Safety: UniqueID(Pet) != Null AppointmentRequest != Null AppointmentConfirmation != Null ReminderData != Null
Permissions	READ: AppointmentRequest, ReminderData WRITE: AppointmentConfirmation, ReminderData CONSUME: AppointmentRequest PRODUCE: AppointmentConfirmation, ReminderData
Activities	ProcessAppointmentRequest, GenerateReminder
Protocols	ScheduleAppointment, SendVetReminder

3. Model Interakcji

Goal	Name	Initiator	Responder	Inputs	Outputs	Description
Pet's vital data	RecordPhysiologicalData	HealthMonitor	PetOwner	EntityID: Pet, Data: PhysiologicalData		HealthMonitor sends physiological data to PetOwner for review and storage.
Pet's location	RecordLocation	LocationTracker	PetOwner	EntityID: Pet, Data: LocationData		LocationTracker sends current location data to PetOwner for tracking.
Pet's behavior	AnalyzeBehavior	BehavioralAnalyzer	PetOwner	EntityID: Pet, Data: BehaviorData		BehavioralAnalyzer sends behavioral insights to PetOwner for awareness.
Pet's health	MonitorHealthStatus	HealthMonitor	VeterinaryClinic	EntityID: Pet, Data: HealthStatus		HealthMonitor sends health updates to VeterinaryClinic for consultation.
Health alert to	SendHealthAlert	HealthMonitor	PetOwner	EntityID: Pet, Data: AlertData		HealthMonitor notifies PetOwner of abnormal health indicators.
Veterinary appointment	ScheduleAppointment	PetOwner	AppointmentScheduler	EntityID: Pet, Data: AppointmentRequest	Data: AppointmentConfirmation	PetOwner requests an appointment with VeterinaryClinic through AppointmentScheduler.
Reminder for vet	SendVetReminder	AppointmentScheduler	PetOwner	EntityID: Pet, Data: ReminderData		AppointmentScheduler sends appointment reminders to PetOwner.

4. Model Usage

Service	Agent	In/Out	Pre/Post Condition	Derived From
RecordPhysiologicalData	HealthMonitoringAgent	IN: EntityID: Pet, Data: PhysiologicalData OUT:	PRE: UniqueID(Pet) != Null POST: PhysiologicalData != Null	RecordPhysiologicalData
RecordLocation	LocationTrackingAgent	IN: EntityID: Pet, Data: LocationData OUT:	PRE: UniqueID(Pet) != Null POST: LocationData != Null	RecordLocation
AnalyzeBehavior	BehavioralAnalysisAgent	IN: EntityID: Pet, Data: BehaviorData OUT: Data: BehaviorData	PRE: UniqueID(Pet) != Null AND BehaviorData != Null POST: BehaviorData != Null	AnalyzeBehavior
MonitorHealthStatus	HealthMonitoringAgent	IN: EntityID: Pet, Data: HealthStatus OUT:	PRE: UniqueID(Pet) != Null POST: HealthStatus != Null	MonitorHealthStatus
SendHealthAlert	HealthMonitoringAgent	IN: EntityID: Pet, Data: AlertData OUT:	PRE: UniqueID(Pet) != Null POST: AlertData != Null	SendHealthAlert
ScheduleAppointment	AppointmentSchedulingAgent	IN: EntityID: Pet, Data: AppointmentRequest OUT: Data: AppointmentConfirmation	PRE: UniqueID(Pet) != Null AND AppointmentRequest != Null POST: AppointmentConfirmation != Null	ScheduleAppointment
SendVetReminder	AppointmentSchedulingAgent	IN: EntityID: Pet, Data: ReminderData OUT:	PRE: UniqueID(Pet) != Null POST: ReminderData != Null	SendVetReminder
ReviewPhysiologicalData	PetOwnerAgent	IN: EntityID: Pet, Data: PhysiologicalData OUT:	PRE: UniqueID(Pet) != Null AND PhysiologicalData != Null POST: PhysiologicalData != Null	ReviewPhysiologicalData
ReviewLocationData	PetOwnerAgent	IN: EntityID: Pet, Data: LocationData OUT:	PRE: UniqueID(Pet) != Null AND LocationData != Null POST: LocationData != Null	ReviewLocationData
ReviewBehaviorData	PetOwnerAgent	IN: EntityID: Pet, Data: BehaviorData OUT:	PRE: UniqueID(Pet) != Null AND BehaviorData != Null POST: BehaviorData != Null	ReviewBehaviorData
ReceiveHealthAlert	PetOwnerAgent	IN: EntityID: Pet, Data: AlertData OUT:	PRE: UniqueID(Pet) != Null AND AlertData != Null POST: AlertData != Null	ReceiveHealthAlert
ReceiveVetReminder	PetOwnerAgent	IN: EntityID: Pet, Data: ReminderData OUT:	PRE: UniqueID(Pet) != Null AND ReminderData != Null POST: ReminderData != Null	ReceiveVetReminder
ConsultPetHealth	VeterinaryClinicAgent	IN: EntityID: Pet, Data: HealthStatus OUT:	PRE: UniqueID(Pet) != Null AND HealthStatus != Null POST: HealthStatus != Null	ConsultPetHealth
ProvideTreatment	VeterinaryClinicAgent	IN: EntityID: Pet, Data: HealthStatus OUT:	PRE: UniqueID(Pet) != Null AND HealthStatus != Null POST: HealthStatus != Null	ProvideTreatment
CollectPhysiologicalData	HealthMonitoringAgent	IN: EntityID: Pet OUT: Data: PhysiologicalData	PRE: UniqueID(Pet) != Null POST: PhysiologicalData != Null	CollectPhysiologicalData
AnalyzePhysiologicalData	HealthMonitoringAgent	IN: EntityID: Pet, Data: PhysiologicalData OUT: Data: HealthStatus	PRE: UniqueID(Pet) != Null AND PhysiologicalData != Null POST: HealthStatus != Null	AnalyzePhysiologicalData
DetectHealthAnomaly	HealthMonitoringAgent	IN: EntityID: Pet, Data: HealthStatus OUT: Data: AlertData	PRE: UniqueID(Pet) != Null AND HealthStatus != Null POST: AlertData != Null	DetectHealthAnomaly
CollectBehaviorData	BehavioralAnalysisAgent	IN: EntityID: Pet OUT: Data: BehaviorData	PRE: UniqueID(Pet) != Null POST: BehaviorData != Null	CollectBehaviorData
AnalyzeBehaviorPattern	BehavioralAnalysisAgent	IN: EntityID: Pet, Data: BehaviorData OUT: Data: BehaviorData	PRE: UniqueID(Pet) != Null AND BehaviorData != Null POST: BehaviorData != Null	AnalyzeBehaviorPattern
CollectLocationData	LocationTrackingAgent	IN: EntityID: Pet OUT: Data: LocationData	PRE: UniqueID(Pet) != Null POST: LocationData != Null	CollectLocationData

Service	Agent	In/Out	Pre/Post Condition	Derived From
ProcessAppointmentRequest	AppointmentSchedulingAgent	IN: EntityID: Pet, Data: AppointmentRequest OUT: Data: AppointmentConfirmation	PRE: UniqueID(Pet) != Null AND AppointmentRequest != Null POST: AppointmentConfirmation != Null	ProcessAppointmentRequest
GenerateReminder	AppointmentSchedulingAgent	IN: EntityID: Pet, Data: AppointmentRequest OUT: Data: ReminderData	PRE: UniqueID(Pet) != Null AND AppointmentRequest != Null POST: ReminderData != Null	GenerateReminder

5. Model Znajomości

